

Identification of a weak muscle group tolerates more joint-angle error than kinematic reliability conventions allow: a predictive-simulation study

Mu-Hua Wang¹, Hui-Ting Shih^{2*}

¹ Department of Biomedical Engineering, National Yang Ming Chiao Tung University, Taipei, Taiwan

² Department of Physical Therapy and Assistive Technology, National Yang Ming Chiao Tung University, Taipei, Taiwan

* Corresponding author. E-mail address: huiting.shih@nycu.edu.tw

Highlights

- Weak-group identification stayed above twice chance at 12° of joint-angle error.
- Tolerance depended on deficit severity more than on which muscle group was weak.
- Moderate weakness was the case most exposed to degraded capture quality.
- Mild weakness was never identified reliably, even at laboratory capture quality.
- Ground reaction force added robust value only from about 8° of error.

Abstract

How much joint-angle error markerless motion capture may carry is usually judged against a convention drawn from a review of kinematic reliability, under which error up to 2° is acceptable, 2–5° needs consideration and error above 5° may mislead. That convention concerns measurement quality and does not appear to have been tested against a clinical decision, so we scored capture quality by diagnostic accuracy instead. Unilateral weakness of one sagittal muscle group was imposed at 20%, 40% and 60% force reduction in forward-dynamic simulations with a Geyer–Herr reflex controller, re-optimised after each imposition, across five whole-body strength variants (66 of 75 design cells). Zero-mean Gaussian error of 0–12° was added to the raw joint angles before any feature was computed, and the weakened group was ranked from sagittal kinematics with and without ground reaction force, leave-one-variant-out. Identification fell from 0.845 at 0° to 0.549 at 12°, remaining at least twice the 0.227 majority-class baseline throughout. Tolerance was not uniform across severity, since at 2° the 40% and 60% deficits reached 0.948 and 0.963 while the 20% deficit reached only 0.610, and degrading capture to 8° cost the moderate case 0.385, about three times either extreme, though five variants cannot separate that difference. Ground reaction force was indistinguishable from zero at 0–2° and added 0.116 at 5°, although priced against the better camera-only feature set that gain falls to 0.033 and kinetics contribute robustly only from 8°. Mild weakness was never identified reliably, so the question a capture specification should answer may be less whether a system is accurate than for which deficit magnitude it is accurate enough.

1. INTRODUCTION

Unilateral muscle weakness after neurological injury tends to be group-specific. The plantarflexors, dorsiflexors, knee extensors, knee flexors and hip flexors each produce a characteristic sagittal gait deviation, and these deviations appear to differ in kind and not only in degree between groups (van der Krogt et al., 2012; Ong et al., 2019). Which group is weak therefore bears on which strengthening programme is indicated. Manual muscle testing provides a coarse and effort-dependent estimate of that, whereas instrumented gait analysis provides a more precise one but is available only in centres that operate a motion-capture laboratory.

Markerless motion capture may remove that constraint. In a six-case series spanning cerebral palsy and chronic stroke, markerless sagittal waveforms matched marker-based ones to within 2.0° in amplitude on the more-affected limb, although agreement was coarser (Steffensen et al., 2023). Smartphone-based capture has been reported to give 4.5° of mean absolute error in ten healthy adults (Uhlrich et al., 2023), and across normal gait plus imitated crouch, circumduction and equinus in 21 healthy volunteers the root-mean-square error was 5.8° with an 11.3° mean peak error, lowest for normal gait and highest for crouch and circumduction (Horsak et al., 2023). The range swept here was chosen to cover the errors these systems report.

How good the measurement has to be is less clear. The working answer is a convention taken from a systematic review of kinematic reliability, under which error of 2° or less is acceptable, $2\text{--}5^\circ$ is reasonable but needs consideration, and error above 5° may be large enough to mislead interpretation (McGinley et al., 2009). That judgement concerns measurement quality and not any particular decision, and it does not appear to have been tested against one. Hardware comparisons we could locate score alternatives by error against laboratory capture and none asks whether the clinical conclusion survives, so whether a system several degrees worse reaches the same decision is not established.

Answering that question requires ground truth, which a patient cannot supply, since the state of each muscle group is not known independently of the assessment being evaluated. In a simulation the weakness is imposed and the capture error added in controlled amounts. Ong et al. (2019) imposed graded plantarflexor weakness in the same reflex-controlled framework used below, and found that severe weakness drives a slow, heel-walking gait with reduced propulsion. Van der Krogt et al. (2012) found gait remarkably robust, several groups tolerating 80% force loss with near-normal kinematics, with the tolerance differing sharply between groups. Together these suggest that a weakness produces a characteristic deviation that scales with severity and differs by group. Related work has simulated unilateral graded weakness (Johnson et al., 2022; Yuan et al., 2026), compensation strategies (Knarr et al., 2013) and finer than group resolution (Park et al., 2023), and shown sensitivity to weakness to depend on the assumed control structure (Kuska et al., 2022), but none asks what follows for measurement.

The purpose of this study was to determine how much joint-angle measurement error the identification of a weak muscle group can tolerate. We used forward-dynamic simulation to impose unilateral weakness of one sagittal muscle group at three severities, added joint-angle error in controlled amounts, and scored capture quality by the accuracy with which the weakened group could be recovered, and not by the measurement error itself. We asked how much joint-angle error that identification can tolerate, whether the tolerance depends on the severity of the deficit or on which group is weak, and whether adding ground reaction force (GRF) extends it.

II. METHODS

A. Simulation and imposed weakness

Gait was generated in SCONE 2.4.4 by forward dynamics with a Geyer–Herr reflex controller (Geyer and Herr, 2010)

on the planar H0914 model (OpenSim 4 engine). Forward dynamics was chosen deliberately, because a tracking formulation reproduces measured unimpaired kinematics, so a converged solution resembles unimpaired gait by construction and the deviation of interest is suppressed. Here the gait is predicted and the deviation is an output, and the limbs are mechanically independent, with no periodicity or mirror constraint, so one-sided weakness can express itself as asymmetry.

Weakness was imposed on the left side only by scaling maximum isometric force for one group at a time: plantarflexors (soleus, gastrocnemius), dorsiflexors (tibialis anterior), knee extensors (vasti), knee flexors (hamstrings) and hip flexors (iliopsoas), at 20%, 40% and 60%. All five act in the sagittal plane, matching the plane the feature set observes, so frontal-plane mechanisms are neither imposed nor claimed. The controller was re-optimised with CMA-ES after each imposition, warm-started from the healthy solution, so that the walker adapts to the deficit. The severity range straddles a published threshold, since generalised weakness of about 40% appears to be tolerated with near-normal kinematics and pathological features emerge above roughly 60% (van der Krogt et al., 2012; Waterval et al., 2021).

Five model variants were used, each a whole-body rescaling of the same musculoskeletal model by a single strength factor: the base model at 1.00 and four rescalings at 0.85, 0.95, 1.05 and 1.10. A variant therefore differs from the others only in how strong every muscle is and is not a separate individual. Crossing five muscle groups, three severities and five variants gives 75 design cells, of which 66 passed quality control. Conditions were rejected if the controller never adapted, if the CMA objective exceeded 5.0, or on pelvis height, distance travelled and trial duration thresholds. The nine unfilled cells, all at 40% or 60% severity, are not interchangeable: one failed to converge, three were not adequately attempted, so they are not evidence of infeasibility, and five passed quality control and were lost to a scripting error (Supplementary S6).

B. Features and capture error

Features were computed on the steady-state window, the last 60% of each trial, the first 40% being discarded as the walker settles from its warm start, leaving about four complete strides per limb. They are sagittal joint-angle range, minimum and maximum per limb, the paretic-minus-non-paretic difference of each, spatiotemporal measures and body-weight-normalised GRF measures, all frame-invariant. The camera tier comprises the kinematic, asymmetry and spatiotemporal features, 33 in total, and the instrumented tier

adds the kinetic ones to give 51. Six of 57 were removed by a rule fixed before any tier comparison, excluding features that are terms of the generating objective (S3, S10).

Error was added to the raw joint-angle signals before any feature was computed. It was zero-mean Gaussian with the stated standard deviation in degrees, drawn independently for each of the 200 samples of each waveform and for each joint and limb, that is, white noise, with four independent realisations per condition per error level. Force and timing measures received a multiplicative error scaling with the same capture quality (1.5% at 2°), so no tier was evaluated undegraded. Error was swept over 0–12°, spanning best-case marker-based measurement through the peak errors reported for markerless systems on non-normal gait patterns (Horsak et al., 2023).

White noise is usually the forgiving case, but for these features it does not appear to be, which is why it is reported as the primary analysis. Range of motion is a difference between extremes and is invariant to the constant offset that dominates real markerless error, whereas white noise inflates both extremes and does move it. A deployment-realistic structure (55% per-trial offset, 30% drift, 15% jitter) gave higher accuracy at every level above 3°, by 0.05 to 0.09, so the white-noise result should be the conservative one for identification accuracy. It is not conservative for the force plate, whose marginal value is smaller under structured error (+0.068 at 8° against +0.163) (S5, S8).

C. Identification and evaluation

Identification was treated as a ranking problem over the five groups using extremely randomised trees (100

estimators, balanced class weights, no hyperparameter tuning), evaluated leave-one-variant-out. Training and testing were at the same capture-error level, and the feature table was rebuilt from the raw signals at each level with the folds cut within it, so each point measures how much information survives that much error and not how a clean-data model transfers to noisy data. Because the groups are unequally represented the reference is the majority-class baseline (top-1 0.227, top-3 0.667). Both tiers were evaluated on the same conditions, folds and error realisations, and the paired difference was bootstrapped over conditions (4000 resamples, 8 seeds). Intervals within a severity stratum (Table 2) were bootstrapped over conditions in that stratum (S11), whereas intervals on the tier difference (Table 4) are t intervals on the five per-variant gains, the variant being the unit that would be redrawn if the study were repeated (S7). With 66 conditions, one condition changing its answer moves any accuracy by 0.015, which we take as an interpretability floor below which differences are reported but not interpreted.

III. RESULTS

A. Simulated deviations

Because the weakness is imposed, the only external anchor available is whether each simulated deviation resembles what the literature reports. Each mechanism was checked in the gait phase to which it belongs, with phases segmented per limb from vertical GRF. Table 1 reports the proportion of conditions in which the expected direction held, both against the contralateral limb and against the variant's own unimpaired solution; the second reference matters because a direction that holds only against the sound limb suggests the asymmetry arose there (S1).

Table 1. Face validity across all conditions of each mechanism, measured in the relevant phase.

Imposed weakness	Expected consequence (phase)	vs non-paretic	vs own unimpaired gait	Verdict
Hip flexors	reduced peak hip flexion (swing)	14/14	14/14	✓
Plantarflexors	reduced peak plantarflexion (terminal stance)	15/15	13/15	✓
Knee extensors	greater peak knee flexion (swing)	10/10	10/10	✓
Knee extensors	reduced knee extension (terminal stance)	9/10	9/10	✓
Dorsiflexors	reduced peak dorsiflexion (swing)	15/15	10/15	✓†
Knee flexors	genu recurvatum (terminal stance)	12/12	8/12	✓*†
Knee flexors	swing knee flexion little changed	11/12	6/12	✓*†

Hip flexors, plantarflexors and knee extensors held on both references. At 60% force loss the paretic ankle reached only -1.1° of plantarflexion in terminal stance against -10.0° on the sound side, exaggerating the reduced, though not absent, toe-off plantarflexion reported after stroke (Kesar et al., 2009). Knee extensor weakness did not produce the naive

expectation, in that the knee did not yield further in loading response (4 of 10). What changed were the two events the vasti control, early-swing flexion (10/10) and terminal-stance extension (9/10), which is consistent with vasti tolerating substantial weakening with near-normal kinematics (van der Krogt et al., 2012). Measured over the whole cycle rather

than inside its own phase, that row reads 9 of 10 instead of 4 of 10 (S1.2). In Table 1 an asterisk marks the two rows on which the literature is divided, a knee-flexor role in knee hyperextension being supported by Okada et al. (2024) but not by Cooper et al. (2012). A double dagger marks the three rows that hold against the contralateral limb but in two-thirds or fewer of conditions against the variant's own unimpaired gait, so part of that asymmetry appears to have arisen on the sound side. Two of those three rows are dose-dependent (S1).

B. Tolerance to capture error

Identification fell from 0.845 at 0° to 0.549 at 12° (Figure 1a), a spread of 0.296, so the measurement had room to degrade before approaching chance. Even at the largest error

tested it remained more than twice the majority-class baseline. The convention places the boundary of usable measurement at 5°, whereas accuracy here was 0.659 at 5° and stayed above 0.5 at 12°. Read as a rate, the curve is steepest inside the band the convention identifies. The loss per degree was -0.011 over 0–2°, -0.055 over 2–5°, -0.020 over 5–8° and -0.013 over 8–12°, with the 2–5° band the steepest in all three rebuilds, although the 2–5° and 5–8° rates are not separable (S14.1). What the data do not support is the step from sensitivity to unusability, since the level at 5° is still 2.9 times baseline and at 12° still 2.4 times. Scoring one capture at a time, without pooling a condition's four noise realisations, lowers the curve but changes no conclusion drawn from it (S14.2).

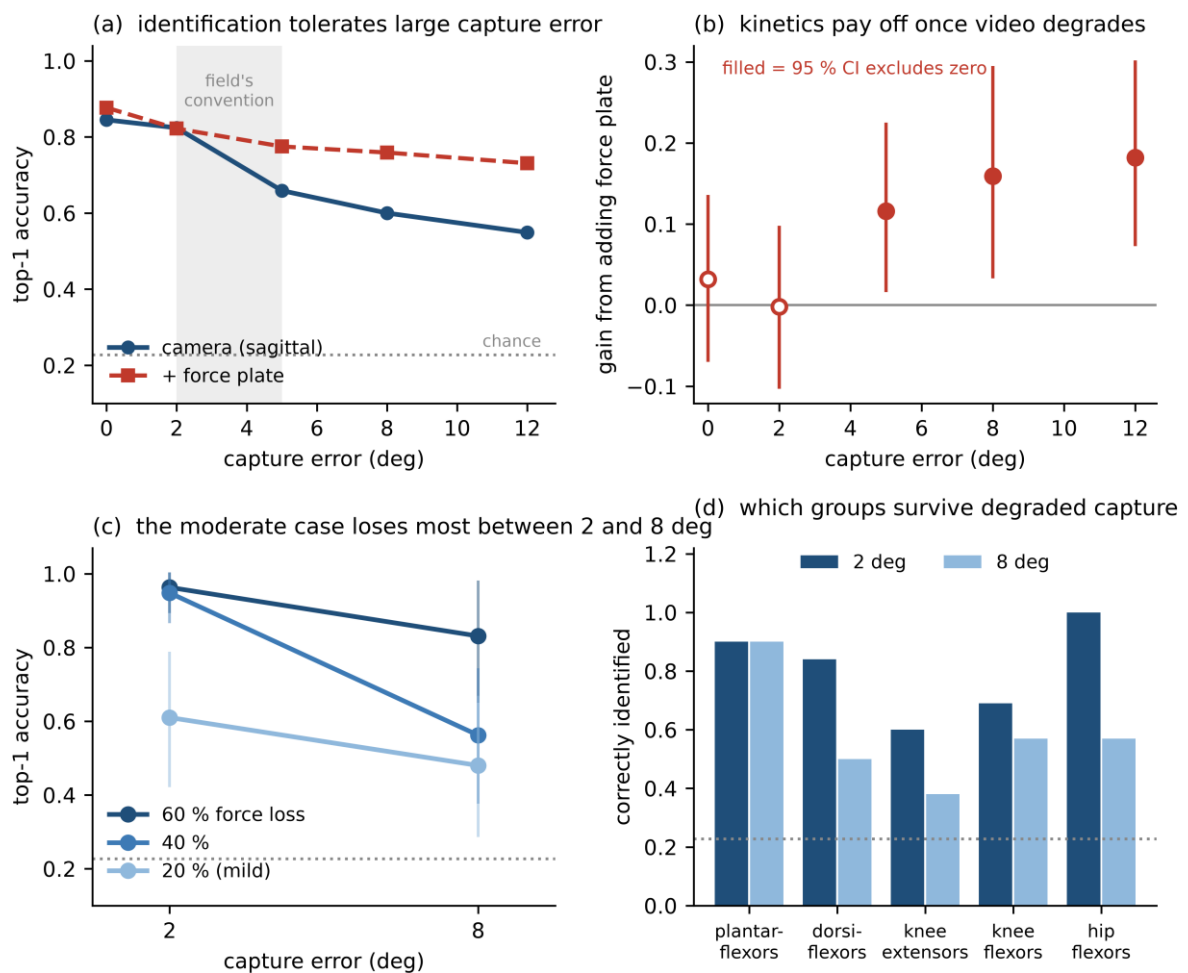


Figure 1. (a) Top-1 identification accuracy against capture error for the camera tier and the camera-plus-force-plate tier; dotted line is the majority-class baseline, shaded band the field's 2–5° convention. (b) Marginal value of adding GRF with 95 % t intervals over the five variants (Table 4); filled points exclude zero. (c) Identification by imposed severity at two capture qualities. (d) Identification by muscle group at two capture qualities.

The result does not appear to be an artefact of unequal optimisation depth between variants. A classifier given only convergence metadata identified the group at 0.178, below the 0.200 chance level, and removing the two most deeply optimised variants lowered the curve at every error level but

left the decline intact, a fall of 32.2% without those two variants against 34.9% with all five (S2, S4).

C. Severity and muscle group

Identification by imposed severity is given in Table 2 and Figure 1c. At laboratory quality, 40% and 60% weakness

were identified almost perfectly and mild weakness was the only difficult case. Degrading capture to 8° cost each severity differently, and not in the order intuition might suggest. The moderate case lost 0.385 against 0.132 for the severe case and 0.130 for the mild one, so it lost more between 2° and 8° than either extreme, by 0.253 and 0.255 respectively, while severe and mild lost indistinguishably from each other. These are means of the per-condition losses, so they need not equal the difference of the rounded accuracies in Table 2. The

Table 2. Identification by imposed severity, top-1 and top-3 accuracy at 2° and 8° of capture error. Intervals are bootstrapped over conditions within each severity (Supplementary S11); n is the number of conditions.

Force loss	top-1 at 2° (95 % CI)	top-1 at 8° (95 % CI)	top-3 at 2°	top-3 at 8°	n
60 %	0.963 [0.897, 1.000]	0.831 [0.654, 0.978]	1.000	1.000	17
40 %	0.948 [0.870, 1.000]	0.562 [0.380, 0.740]	1.000	0.958	24
20 %	0.610 [0.425, 0.785]	0.480 [0.290, 0.665]	0.970	0.855	25

Degraded capture does not appear to be what makes mild weakness hard to identify, since it was already the hardest case at laboratory quality and better equipment did not recover it. Top-3 accuracy behaved differently, the correct group being in the top three in 97% of mild cases at 2° though ranked first in only 61%. The same interaction is present by muscle group (Figure 1d). Plantarflexor weakness was identified at 0.90 at both 2° and 8°, while dorsiflexors fell from 0.84 to 0.50, knee extensors from 0.60 to 0.38, knee

direction held in four of the five variants, but with only five variants the difference is not statistically separable, and about a quarter of the moderate-versus-severe gap may be the unbalanced mechanism mix and not severity itself (S11). The mild-weakness figure is the least stable in the study, at 0.799, 0.650 and 0.610 across three independently regenerated datasets (S4), so no conclusion is drawn that depends on which end of that range is correct.

flexors from 0.69 to 0.57 and hip flexors from 1.00 to 0.57. The errors do not appear to be random (Table 3), since the dominant substitutions at 2° were knee extensors called plantarflexors (22%), knee flexors called hip flexors (18%) and dorsiflexors called hip flexors (13%), pairs sharing a role in stance support or limb advancement, and the same pairs dominated at 8°. A clinician would therefore need to know which substitutions such a system makes and not only how often it is right.

Table 3. Confusion at 2° of capture error. Rows are the imposed weakness, columns what the system identified, in per cent.

Imposed	PF	DF	KE	KF	HF
Plantarflexors	90	1	6	3	·
Dorsiflexors	·	84	·	2	13
Knee extensors	22	10	60	8	·
Knee flexors	·	14	·	69	18
Hip flexors	·	·	·	·	100

D. Kinetics

The marginal value of adding GRF to the reported camera tier (Figure 1b) was +0.032 at 0° and -0.002 at 2°, both indistinguishable from zero, then +0.116 at 5°, +0.159 at 8° and +0.182 at 12°, with intervals given in Table 4. Because the variant is also the cross-validation fold,

conditions within a variant are not independent, and the spread between them is large, the five variants scoring 0.644 to 0.973 at 2°, a wider range than separates any two capture qualities (S2). Table 4 therefore takes the variant as the unit. The gain was positive in all five variants at 5° and above and mixed in sign below it.

Table 4. Marginal value of ground reaction force. Intervals are *t* intervals on the five per-variant gains, the variant being the unit that would change if the study were repeated. They are centred on the unweighted mean of the five, which differs from the condition-weighted gain in the third decimal.

Capture error	Gain	Per-variant gains	95 % CI over the five variants	Excludes zero
0°	+0.032	-0.107 to +0.098	[-0.068, +0.134]	no
2°	-0.002	-0.116 to +0.087	[-0.101, +0.096]	no
5°	+0.116	+0.045 to +0.239	[+0.018, +0.223]	yes
8°	+0.159	+0.027 to +0.295	[+0.035, +0.293]	yes
12°	+0.182	+0.054 to +0.307	[+0.075, +0.300]	yes

What the camera tier contains moves the useful crossover from 5° to 8°. Six spatiotemporal features survive the objective-term exclusion, and in this model all derive from vertical GRF, which a camera alone would not supply. The remaining 27 purely kinematic features score 0.824, 0.875, 0.742, 0.623 and 0.536 across the sweep, and are in fact the better camera-only configuration at 2°, 5° and 8°. Because that subset is free, the honest baseline for a force plate is

whichever camera-only tier is better, and against it the plate adds +0.032, -0.053, +0.033, +0.136 and +0.182, so the correction bites hardest at 5°, where the gain falls to +0.033, barely twice the interpretability floor. The 33-feature tier is retained as the headline because it differs from the instrumented tier by the force plate alone. Where the spatiotemporal features are assigned moves the crossover the same way, the 5° gain being +0.110 and +0.102 under the

two assignments that treat timing as camera-derived but +0.019 under the conservative assignment (S3). The 8° and 12° gains survived all three assignments, none falling below +0.110 and +0.186, though these are separate six-seed runs and so bracket the reported +0.159 and +0.182 without including them (S3, S4). Kinetics therefore begin to contribute at 5°, and 8° is the first level at which the contribution appears robust both to how the tiers are drawn and to what a camera can actually deliver. We report 8° as the level this design supports and not as a purchasing threshold, since with five variants, and between-variant variation larger than the effect itself, the ordering is firmer than the number.

The dataset was rebuilt twice, once to add conditions and once to repair a provenance defect, so every headline was measured three times on independently regenerated data (Table 5). The direction of all three findings held each time: accuracy declined with capture error and stayed above baseline, severe weakness was identified better than mild, and kinetics contributed at high capture error and not at low. Magnitudes moved, mild-weakness accuracy at 2° most, and the one quantity that changed sign is the marginal value of kinetics at 0–2°, reported here only as indistinguishable from zero (S4).

Table 5. Every headline measured on three independently regenerated datasets. The provenance-repaired 66-condition rows are the study's result; the other two show how far the magnitudes moved (Supplementary S4).

Dataset	0°	2°	5°	8°	12°
<i>Identification, camera tier</i>					
61 conditions, stale provenance	0.891	0.842	0.758	0.689	0.617
71 conditions, first rebuild	0.840	0.794	0.710	0.674	0.655
66 conditions, provenance repaired	0.845	0.824	0.659	0.600	0.549
<i>Marginal value of the force plate</i>					
61 conditions, stale provenance	−0.018	−0.023	+0.027	+0.061	+0.076
71 conditions, first rebuild	+0.032	+0.040	+0.077	+0.104	+0.095
66 conditions, provenance repaired	+0.032	−0.002	+0.116	+0.159	+0.182

IV. DISCUSSION

Identification of a weak muscle group tolerated considerably more joint-angle error than the convention used to judge capture quality would allow, accuracy at 12° remaining more than twice the majority-class baseline, so as a limit on usability that convention appears stricter than this particular decision requires. Tolerance depended more on the severity of the deficit than on which group was weak, and ground reaction force extended it only once kinematic quality had degraded well past the convention's own boundary. The steepest degradation per degree did fall inside the $2\text{--}5^\circ$ band in every rebuild, although no interval separates that band from the one adjoining it. A threshold might therefore usefully state which decision it protects, and whether it marks where accuracy falls fastest or where it stops being sufficient.

Severe weakness was identified reliably whatever the capture quality, whereas mild weakness was not, at any quality tested, and the relevant figure there may be the 0.970 top-3 at 2° rather than the 0.610 top-1, since the method can shortlist a mild case without resolving it. That shortlist is itself capture-dependent, falling to 0.855 by 8° , so it appears to be a laboratory-quality fallback and not a robust one. The case that capture quality appears to determine is the moderate one, which fell from 0.948 to 0.562 between 2° and 8° , the largest fall of the three though not separably so.

The two studies that bracket this problem suggest a mechanism. Gait appears to absorb large force losses with near-normal kinematics (van der Krogt et al., 2012), while the characteristic deviation emerges only once weakness is severe (Ong et al., 2019). A severe deficit may therefore produce a signature large enough to survive substantial measurement error, and a mild one a signature too small to read even at laboratory quality, with the moderate deficit sitting where the deviation and the measurement error are of comparable size. The clinical reading is consequently narrower than an instruction to buy a better camera to catch subtle weakness. For mild deficits equipment quality does not appear to be the binding constraint, whereas for moderate deficits it may be. This could interact unfavourably with a reported property of markerless systems, since error was lowest for normal gait and highest for crouch and circumduction, at least in healthy volunteers imitating them (Horsak et al., 2023), so if that ranking carries over to patients, moderate deficits would be measured less reliably than severe ones.

At laboratory capture quality the force plate contributed nothing detectable, and appears to become worth its cost only

once kinematic quality degrades badly, at 8° and beyond. A similar null has been reported in a clinical population, since in classifying cerebral-palsy gait patterns kinematics considerably outperformed ground reaction force (93.4% against 47.2% peak accuracy) and adding force gave no advantage (Slijepcevic et al., 2023). We propose pipeline independence rather than noise immunity as the mechanism, because a force plate does not inherit kinematic error whereas kinetics derived from a musculoskeletal model do, so substituting inverse-dynamic moments should not reproduce this.

The objection that the classifier merely recovers what was injected is really two questions. Waterval et al. (2021) validated this framework against sixteen patients with bilateral plantarflexor weakness, reporting $R > 0.64$ for hip and ankle kinematics and kinetics but $R < 0.55$ for knee kinetics. The half that agrees least is the knee, which is also where our own face validity is weakest and where our worst confusions sit, so the two weaknesses are plausibly the same one. Whether the inverse step reads a deviation or a simulator fingerprint is testable here, because a fingerprint would be destroyed rather than degraded by added error, and the off-diagonal confusion pattern at 2° predicts it at 5° , 8° and 12° at $\rho = 0.65$, 0.73 and 0.55 , each above a permutation null (S12). Magnitude remains unaddressed, since no deviation here has been matched to a patient's.

This study has several important limitations. It is entirely in-silico, and establishes what is identifiable in principle from a model with known ground truth, and does not establish clinical accuracy. The unimpaired simulation is slower, lower in cadence and larger in hip range than a healthy adult, a known property of planar reflex-controlled walkers, and the severity axis is not a clinical severity scale (S9). The model is planar, excluding any frontal-plane mechanism, and its reflex controller is bilaterally parameter-shared, so the imposed asymmetry is mechanical and the neural asymmetry accompanying unilateral injury is not represented. The five variants are strength rescalings of one model, so leave-one-variant-out tests generalisation across strength rather than across people, and five is a small number of clusters for the intervals in Table 4. The grid is incomplete, at 66 of 75 cells, the missing ones at higher severities. Capture error was modelled on joint angles and not propagated through a capture pipeline, and a nominal σ induces only $0.4\text{--}0.9\sigma$ of perturbation in a feature, so the degree axis is not interchangeable with a reported root-mean-square error (S5, S8.1). Finally, no unimpaired condition was included, so detection of weakness, as distinct from identification of its location, was not tested.

The 8° figure does not license a claim that any particular clinical setting is adequate. Published markerless validation is almost entirely laboratory-based, and no study we could locate reports the joint-angle error of a capture volume the size of a real clinic. What the 8° figure identifies is the measurement that is missing and the value it would have to meet.

Conflict of interest

None declared.

Data availability

The analysis scripts listed in Supplementary S13, the quality-control record and the frozen feature cache from which every reported number is computed are provided as supplementary files and deposited at [repository DOI to be inserted].

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CRediT authorship contribution statement

Mu-Hua Wang: conceptualisation, methodology, software, formal analysis, investigation, writing — original draft. **Hui-Ting Shih:** conceptualisation, supervision, writing — review and editing.

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One scripting fault occurred during the study, with two consequences: a monitoring script restarted optimisations after the dataset had been frozen, deleting five cells that had passed quality control (§2.1) and displacing the stored trajectories of ten further conditions, which were re-evaluated before analysis; the re-evaluated trajectories are what the reported results use (Supplementary S6). The analysis scripts, the quality-control record and the feature cache from which every reported number is computed are provided.

Figure legend

Declaration of generative AI and AI-assisted technologies

Anthropic's Claude was used beyond language editing: it implemented and ran the analysis scripts listed in Supplementary S13, produced Figure 1 from their outputs, and drafted and revised the text under author direction. The authors set the research questions, decided which claims to make, and are responsible for the content. Every reported number is produced by a named script operating on the frozen dataset, both provided, so any figure here can be recomputed independently of the tool that generated it. No

AI tool is listed as an author, and no image was generated or altered by AI.

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